Web Service Recommendation via Integrating Heterogeneous Graph Attention Network Representation and FiBiNET Score Prediction

Buqing Cao, Mi Peng, Lulu Zhang, Yueying Qing, Bing Tang, Guosheng Kang, and Jianxun Liu

***Abstract*—The rapid growth in the number and diversity of Web service, coupled with the myriad of similar Web service in functionality, makes it challenging to find most suitable Web service for users to accelerate and accomplish Mashup development. Therefore, this article proposes a Web service recommendation method via integrating heterogeneous graph attention network representation and FiBiNET (Feature Importance and Bilinear feature Interaction NETwork) score prediction. In this method, first, a heterogeneous information service network is constructed by using composite service information, atomic service information, and their respective attribute information. Second, the meta-paths are defined according to different semantic information and service similarity matrixes are built by using commuting matrix and meta-path-based similarity measurement technology. A two-layer attention model is designed to calculate the node level attention and meta-path-level attention of the services respectively, and generate the feature representation of Web service. Third, for the Web services in the service cluster, combining their feature representations with multi-dimensional QoS attributes, the FiBiNET is exploited to dynamically learn the importance of features and complex feature interactions, and predict the score of Web services. Finally, the experiments are performed on the real Web service dataset. The experimental results show that the proposed method is better than the other nine methods in terms of accuracy, recall, F1, and AUC, and achieves better classification and recommendation quality.**

***Index Terms*—Service recommendation, heterogeneous information network, meta-path, attention mechanism, FiBiNET.**

I Introduction

MASHUP is a typical representative of Web 2.0 technology. As an application development technology for integrating data, Mashup can integrate relevant information very quickly to meet the actual needs of situational applications [1]. For example, Google Maps offers Web APIs that support maps and YouTube video data, and Amazon provides a fully hosted service that helps developers easily publish, maintain, monitor, and run Web APIs of any size. The emergence of many open platforms enables us to combine the vast amount of Web services in a flexible and orderly manner. Mashup developers can create useful applications. It greatly expands the scope and depth of Web applications. The application of open platforms such as Google Maps in weather, traffic, travel, and other situations shows the potential and value of Mashups.

With the popularity of Web applications, the number of Web services on the internet grows exponentially. As of Jan 2022, the number of Web services counted on the ProgrammableWeb has exceeded 31000. Subsequently, it is becoming a challenge to accurately and quickly find the most suitable Web services from Web services repository with vast amount to create Mashup application. Similar to traditional text information retrieval, Web service discovery is generally achieved by performing similarity matching between users' query and the description of Web services. However, the heterogeneity of type, complexity of the relationship in Web services make service discovery difficult [2]. To solve this problem, some researchers exploit service recommendation techniques to achieve effective service discovery, which mainly include content-based, QoS-based, and hybrid service recommendations [3]. Among this, content-based service recommendation [4], [5] has obvious limitations in accuracy, because the brief description text lacks some detailed available information of Web services. QoS-based service recommendation [6], [7] do not significantly improve their accuracy due to the unstable and difficulty to obtain historical service usage data.

Recently, it is becoming a hot topic to apply deep neural network for hybrid Web service recommendation. They automatically learn the interaction between services to obtain strong feature representation, which can greatly facilitate service recommendation [2], [8]. For example, a Web service recommendation method based on graph convolution neural network is proposed. It describes the word co-occurrence and relationship between documents, and constructs a graph network of heterogeneous words and Web service description documents [8]. Some researchers construct the structure relationship graph and attribute bipartite graph. Then, random walk is used to generate the structure and attribute contexts of Web services. The representation vector of multi-dimensional information can be obtained by training the model [9]. Most of the above works is to model the service information network as a homogeneous information network. However, if the service information network only considers a single type of service node, and does

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not distinguish the heterogeneity of the relationship between the service and other types of nodes (Mashup, tag, etc), it may cause irreversible information loss. Heterogeneous Information Network (HIN) has been widely used in many data mining tasks. The rich Web service information and attribute information contained in service network, as well as different types of relationships formed by the information, are naturally constructed as HIN [10]. By using rich semantics in HIN, the interaction between services can be deeply mined to help the model better learn the similarity between services. At the same time, the emergence of attention mechanism can well cope with the huge challenges brought by the heterogeneity of service network and rich semantics [11]. In other words, more accurate embedding representation for service can be achieved from the diversified auxiliary information in HIN.

Furthermore, in hybrid Web service recommendation, the composition of different features will have a great impact on recommendation quality. Some typical deep neural network models, such as FNN (Feed-forward Neural Network) [32], DeepFM (Deep Factorization Machine) [33], xDeepFM (eXtreme Deep Factorization Machine) [34], are used to mine the low-order and high-order feature interactions of Web service. However, they generally exploit inner product or Hadamard product of feature vector to calculate the feature composition, and often ignore the importance of the features themselves. So, it is difficult in hybrid Web service recommendation to learn the important features that never or rarely appear from the training samples [13]. Latterly, FiBiNET is presented for click-through rate prediction [35]. On one hand, it dynamically learns the importance of features themselves by using the SENET (Squeeze-Excitation network) [35]. On the other hand, it uses a bilinear function to better model the feature composition and interaction. That is to say, the importance of features and feature interactions can be mined and learned to facilitate service recommendation.

To solve the above problems, extended from our previous work [12], i.e., heterogeneous graph attention network-enhanced Web service classification, by combining score prediction based on feature importance and composition, this paper proposes a novel Web service recommendation. It first builds a heterogeneous service information network and refines the meta paths according to different semantics. Second, a two-layer attention model is designed to calculate the node-level attention and meta path-level attention of service, to generate more representative embedding features for achieving more accurate service classification. Third, for the services in the identified service cluster, FiBiNET is applied to dynamically learn the importance of features and complex feature interactions by modeling services' feature representations and multi-dimensional QoS attributes. Finally, the usage probability of services is predicted and ranked, and services with high-quality will be recommended to create Mashup.

In summary, the contributions of this paper are as below:

- A heterogeneous graph attention network-based Web service representation and classification is presented. The heterogeneous service information network makes the relationship between services more reserved, which is of great significance to measure the importance of neighbor services on service nodes that need to be classified. The similarity between Web service can be measured more accurately and the accuracy of service classification can be improved by combining service commuting matrix under different meta-paths to learn the parameter weight of service node features.
- A novel Web services recommendation is designed. Web services' feature representations and multi-dimensional QoS attributes are used as data source of score prediction. FiBiNET is used to dynamically learn the importance of features and complex feature interactions to improve the quality of service recommendation.
- A series of experiments are performed on real datasets crawled from ProgrammableWeb. Experimental results show that compared with other mainstream methods, the proposed method has better classification accuracy and recommendation quality.

The rest of this paper is organized as follows: Section II is the related work. Section III introduces the preliminaries. Section IV is the proposed method. The experimental results are reported and analyzed in Section V. Section VI is the conclusion and future work.

II Related Works

Web service recommendation method can be divided into content-based, quality of service (QoS)-based and hybrid service recommendation.

A. Content-Based Service Recommendation

Content-based service recommendation is modeled by extracting the feature words of service description. The inconsistency of Web service description documents leads to the diversity of Web service content modeling. At present, there are structured documents based on WSDL and service documents based on natural language description, mainly using technologies related to natural language processing, such as TF-IDF [14], Keyword [15], [16], topic model [17], [18], and other technologies. For example, Chen et al. [15] presents a keyword-driven service recommendation via deep reinforced steiner tree search. Wu et al. [15] proposes a hybrid Web services tag recommendation strategy called WSTRec. It first applies tag co-occurrence, tag mining, and semantic correlation measurement techniques to tag recommendation. Then, it presents a novel Web services clustering called WTCluster by comprehensively considering WSDL text and tag [17]. Li et al. [18] uses the LDA topic model to obtain the implied functional characteristics of services from WSDL documents. In our previous work [19], Mashup service description, Web APIs description, and their labels are integrated as clustering data sources, and a Mashup service clustering based on the LDA topic model is proposed. By locating user requests to specific service clusters, the search space and scope of services can be effectively reduced, and the efficiency and accuracy of service discovery can be effectively improved. Most of the above work uses LDA model to perform topic-oriented service recommendation, which improves the accuracy of service recommendation to a certain extent but lacks the deeply mining the relationship among services. In fact, network embeddings [20] can represent nodes in the network as low-dimensional,

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real-valued, and dense vectors by learning node feature vectors and network structure, and the vectors obtained have the ability of representation and reasoning in vector space [21]. So, some network embeddings, such as graph attention network, can be applied to the mining of meth-path or service relationship network for Web service recommendation [22].

B. QoS-Based Service Recommendation

QoS-based service recommendation is to provide users with Web services that may meet their needs by using the historical interactive data of users. Most of them exploits collaborative filtering to recommend information that users are interested in by taking advantage of the preferences of groups with similar interests or common experiences. Zheng et al. [6] employs collaborative filtering to predict QoS and proposes a user collaboration mechanism, which can collect useful QoS of services from user behaviors of different services. Some works [7], [22] introduce the geographic location of users and services and construct a regional model to predict QoS missing values, to improve the accuracy of Web service recommendation. Zhang et al. [24] designs a privacy-preserving QoS forecasting in mobile edge environments. Chang et al. [25] proposes a graph based QoS prediction for Web service recommendation. Gao et al. [26] presents an API recommendation for Mashups based on popular ordering, using Mashup descriptions, API descriptions, and the historical call relationship between Mashups and APIs. The APIs it recommends are not only similar in their Mashup capabilities but also popular and similar to each other. Li et al. [3] designs a topic adaptive Web API recommendation via integrating multi-dimensional information. It uses the similarity between Mashups, the similarity between Web APIs, Web API popularity, and co-occurrence as the inputs of the factorization machine to recommend Web APIs for developers to create Mashup. The advantage of these approaches is to have accurate, true user history data. Meanwhile, they mostly use matrix factorization, factorization machine, and other technologies to recommend Web services, which can alleviate data sparsity and cold start, and so improve the accuracy and efficiency of recommendation. However, the feature composition used in many current works mainly use inner product or Hadamard product of feature vectors to calculate cross or composite features [13]. They are relatively simple, and it is difficult to effectively model feature crossing on service sparse datasets. FiBiNET, which has the SENET mechanism, uses bilinear functions to learn the intersection of features. It uses SENET to consider the importance of different features for prediction. And different features are weighing according to their importance, which can greatly improve the recommendation accuracy.

C. Hybrid Service Recommendation

Hybrid service recommendation combines content-based and QoS-based service recommendation to improve the accuracy of recommendation. Thus, it has become a research hotspot of service recommendation at present. Xia et al. [27] proposes a category-aware API clustering and distributed recommendation, which uses an extended K-means clustering to cluster APIs, and then develops a distributed machine learning framework to predict service ranking. Lin et al. [43] uses the attention-based mechanism to improve the early work recommendation system of Web-based knowledge sharing, which can effectively distinguish the differences between users. Cao et al. [2] presents an API service recommendation method by integrating SOM functionality clustering and DeepFM quality prediction. In addition, Cao et al. [28] combines bilinear graph representation and xDeepFM quality prediction for Web service recommendation. These methods unite service content and QoS and extend to service recommendation based on classification or clustering. This kind of method can make good use of high-precision embedding of service classification to improve the accuracy of service recommendation. Therefore, we use heterogeneous graph attention network [12] to mine the importance of service nodes and meta-path semantics in heterogeneous service information network. And service feature representation and classification are performed by considering the importance of neighbor services on current service nodes. Compared with previous work, the proposed service classification mines the heterogeneous information network of Web services more deeply, extracts the meta path of specific semantics, and enriches sparse service content in previous work. Meanwhile, FiBiNET [29] dynamically learns the importance of features and complex feature interaction to greatly improve the effectiveness of service recommendation.

TABLE I
SYMBOL SYSTEM

Symbol	Implication
G	Heterogeneous Service Information Network (HSIN)
V	The node in HSIN
E	The edge in HSIN
Φ	Node type mapping function
Γ	Edge type mapping function
O	The different object type of node
R	The different relationship type of edge
S	A set of type objects
P	The meta-path in HSIN
M	The commutation matrix of meta-path
W	The adjacency matrix between type objects

III PRELIMINARIES

Traditional service networks only consider representational aggregation between services and neighbor services, which usually ignores the influence of semantic relationship. In this paper, the service embeddings process is enhanced by using the information of service itself and the meta-path-guided neighborhood. In this section, we will introduce the concepts of heterogeneous service information network and other related information. For the sake of the easement of description, the main insignias used below are enumerated in Table I.

Definition 1. Heterogeneous Service Information Network: The information network is defined as a directed graph $G = (V, E)$ with node type mapping function $\Phi : V \to O$ and edge type mapping function $\Gamma : E \to R$. When there is more than

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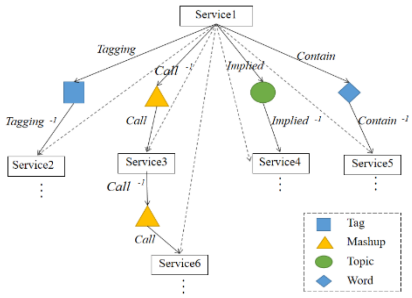

Fig. 1. The relationship between different types of nodes in heterogeneous service information networks.

TABLE II
SEMANTIC OF META-PATH

Notation	Meta-path	Semantics
ATA	API-Tag-API	APIs with the same Tag
AMAMA	API-Mashup-API-Mashup-API	APIs called by Mashup that has called the same API
ADA	API-Topic-API	APIs with the same Topic
AWA	API-Word-API	APIs with the same Word
MTM	Mashup-Tag-Mashup	Mashups with the same Tag
MAM	Mashup-API-Mashup	Mashups calling the same API
MDM	Mashup-Topic-Mashup	Mashups with the same Topic

one type of node or edge, the network is called HIN. Therefore, we define the **H**eterogeneous **S**ervice **I**nformation **N**etwork (HSIN) $G_{SR} = (V_S, E_R)$. $S = \{A, M, T, D, W\}$ includes objects A (Web API), M (Mashup), T (Tag), D (Topic), and W (Word). The edge relation type R includes the tagging relation between services and tags, the calling relation between services and Mashups, the implicit relation between services and topic words, and the inclusion relation between service and feature words. For example, the relationship $R(V_A, V_T)$ indicates that object A has referenced object T, while relationship $R(V_A, V_D)$ indicates that object A contains information obtained by using topic model on the topic D. HSIN is shown in Fig. 1.

Web services in HSIN are connected by different types of relationships, and these paths imply different semantics [29]. Therefore, the meta-path [30] is used to represent the different semantic relations between linked objects.

Definition 2. Neighborhood Guided by Meta-Path: Given the object O and symmetric element path P in HSIN (which means that the type of starting node of the path is the same as the end node's type), we take service node A as the target object need to classify. The meta-path P represents the path obtained by a series of relationship constraints: $A_1 \xrightarrow{R_1} A_2 \xrightarrow{R_2} \ldots \xrightarrow{R_3} A_{l+1}$. The meta-path-guided neighborhood is defined as the set of other services of the same service type A obtained by service type A walking along the given meta-path P. In Table II, we show some symmetric meta-paths defined and their semantics.

Definition 3. Commuting Matrix: For HSIN with the Definition 1, the commutation matrix of meta-path $P = (V_{S_1} V_{S_2} V_{S_3})$ is defined as $M = W_{V_{S_1} V_{S_2}} W_{V_{S_2} V_{S_3}}$. Where $W_{V_{S_i} V_{S_j}}$ is the adjacency matrix between type object S_i and type object S_j.

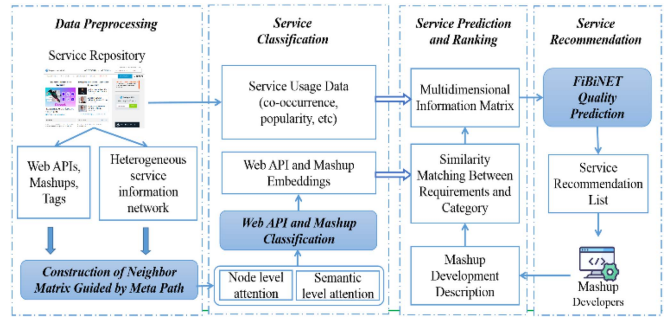

Fig. 2. The overall framework of the proposed method.

The elements in the adjacency matrix are the path instances corresponding to S_i and S_j obtained by meta-path linking [31].

IV METHOD

The overall framework of the method proposed is shown in Fig. 2, which mainly includes the below parts.

- *Data preprocessing:* The service dataset is preprocessed to obtain meta information such as API description text, Tags, and Mashups. For the description text, the feature is processed by word segmentation and the removal of stop words, and the service input feature matrix is obtained by one-hot code. Meanwhile, the meta-paths with specific meanings are extracted by constructing heterogeneous service information network, and the specific commuting matrix is calculated according to these meta-paths.
- *Heterogeneous graph neural Network enhanced service classification:* Taking the initial feature matrix and several specific interaction matrices of meta-paths as input, the self-attention mechanism is used to aggregate neighbor node features for interaction matrices under different meta-paths. The semantic embeddings of node-level service embeddings under each meta-path are obtained by attention mechanism again, and then service classification is performed by Softmax.
- *Service prediction and ranking based on FiBiNET:* For Web APIs in specific clusters, service information matrix is constructed by using Web API/Mashup similarity matrix, popularity, co-occurrences and other QoS attributes. FiBiNET is used to learn the importance of features and complex feature interaction, and predict the recommendation score of Web APIs.
- *Web service recommendation:* To meet Mashup needs of developer users, we calculate the functional similarity between Mashup clusters and services to find the best matching service cluster and recommend high-quality Web APIs for Mashup in this service category.

The details of proposed method are presented as below.

A. Data Preprocessing

After crawling Web services from the Internet, we perform preprocessing to extract useful information of each Web service document. The specific operations are as follows:

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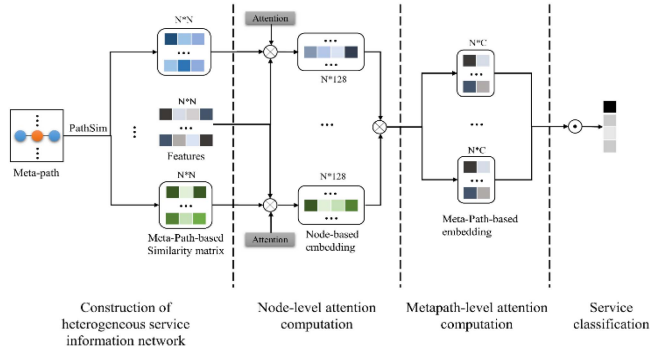

Fig. 3. The framework of Web service classification.

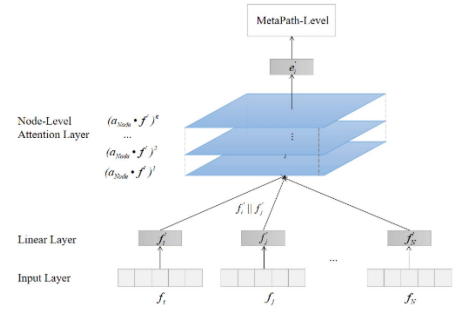

Fig. 4. Node-level attention computation.

1) *Remove punctuation:* Lowercase all the words in the description document to facilitate subsequent word stemming. At the same time, the punctuation in the text is removed, leaving only meaningful words.
2) *Word segmentation:* Filter out the functional words in the language, remove the stop words in a sentence, and then list the remaining words. Each element is the word after removing the stop words.
3) *Word stemming:* We implement word stemming for description document. For example, the root of *creating* and *created* is *create*.

B. Web Service Classification

The basic idea of the Web **S**ervice **C**lassification based on **H**eterogeneous Graph **A**ttention **N**etwork (**SCHAN**) is to design a two-layer attention model based on the meta-path-guided neighborhood to enhance the service representation. In this paper, heterogeneous service information network is constructed by using multiple service relationships, the meta-path is used to guide the link of services, and the graph attention network is employed to capture the neighborhood information. At the same time, the attention mechanism is exploited again to obtain rich semantic information for the aggregation results of different meta-paths, and finally, the service embedding is achieved by integrating rich node information and relationship information. The overall framework of the method is shown in Fig. 3. It will be described in detail in the following sections.

1) Constructing Service Similarity Matrix Based on Meta-Path: It is necessary to distinguish the subtle differences between neighbors and choose some informative neighbors, especially when different node embedding plays different roles and shows different importance. The construction of the similarity matrix is an important part of the proposed model. For each meta-path, PathSim is used to calculate the similarity between services. Given a symmetric meta-path P, $S(A_i, A_j)$ is defined in two ways: the relationship between two services A_i and A_j defined on the meta-path, and the number of path instances between them and themselves. Therefore, the PathSim values of the two services A_i and A_j are shown in Formula (1).

$$S(A_i, A_j) = \frac{2 \times \left|\rho_{A_iA_j} : \rho_{A_iA_j} \in P\right|}{\left|\rho_{A_iA_i} : \rho_{A_iA_i} \in P\right| + \left|\rho_{A_jA_j} : \rho_{A_jA_j} \in P\right|} \tag{1}$$

Where $\rho_{A_iA_j}$ is the path instance between A_i and A_j. When the number of intermediate nodes in the path exceeds 1, the commuting matrix is used to connect the different relationship matrices in the path, as shown in Formula (2).

$$M = W_{R_1} W_{R_2} W_{R_3} W_{R_4} \tag{2}$$

Where W_{R_i} is the relation matrix obtained by the relation R_i between the service and other nodes with other types. M_{ij} is the corresponding value in the commuting matrix of services A_i and A_j in a specific relationship.

Thus, the PathSim values of A_i and A_j evolve into the calculation of the values in the service commuting matrix, as shown in Formula (3):

$$s(A_i, A_j) = \frac{2M_{ij}}{M_{ii} + M_{jj}} \tag{3}$$

After calculating the similarity between two services in each meta-path, the similarity threshold is achieved by setting super parameters. The neighbor services whose similarity exceeds the threshold are identified as neighbor services, and one-hot coding is applied to them. Each adjacency matrix represents a neighborhood based on meta-path guidance.

2) Node Level Attention: Since each meta-path generates an adjacency matrix, we design node level attention calculation, whose input is the feature vector of service, and output is the embeddings vector of service under different meta-paths. It can be shown in Fig. 4.

First, the linear layer is applied to enhance the input representation. Because the initial features are obtained by one-hot code with the word library from the service description text, the number of feature words is tens of thousands, and the feature vector of each service is very sparse. Therefore, during the training process, we use the linear layer to enhance the features, to a certain extent and reduce the sparsity of input features and the burden of model training. It can be shown in Formula (4).

$$f' = W_L F \tag{4}$$

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Where W_L is the parameter matrix of random initialization of linear layer. F is the original characteristic matrix of service, with $F = \{f_1, f_2, \ldots, f_N\}$; is the linear transformation characteristic matrix of service, with $F' = \{f'_1, f'_2, \ldots, f'_N\}$.

Then, we calculate the attention weight of A_i to service A_j according to the adjacency matrix of different meta-paths. Given a service node pair (A_i, A_j) connected by meta-path, all the node pairs based on meta-paths share the parameter vector α. It can be seen from Formula (5) that the weight of node pairs depends on the characteristics of the service itself. Meanwhile, the calculated attention is asymmetric, that is, the importance of service A_i to service A_j is not necessarily the same as that of service A_j to service A_i. This is because the sequence of feature concatenation of service node pairs (A_i, A_j) and (A_j, A_i) is different.

$$a^p_{nodeij} = soft\max(\sigma(\alpha^T_p \bullet [f'_i || f'_j])) \tag{5}$$

Among them, f'_i is the enhanced feature obtained by a layer of linear transformation, α^T_p is the shared parameter vector of node-level attention layer under the meta-path p, and the parameter is trained by a layer of feedforward neural network. $||$ stands for concatenation operation, which avoids the increase of training parameters. σ is a LeakyRelu nonlinear activation function. Finally, the value a^p_{nodeij} obtained by Softmax normalization is the attention coefficient of neighbor service A_j to service A_i under p.

Next, the attention weights and features of all neighbor nodes in each meta-path are summed to obtain the node embeddings based on the meta-path. As shown in Formula (6), each service node is embedded by its neighbor service aggregation. Since the node level attention weight is obtained under a single meta-path, it has the semantic specificity attached to the meta-path.

$$e^p_i = \sigma\left(\sum_{j \in Ne^p_i} a^p_{nodeij} \bullet f'_j\right) \tag{6}$$

Where Ne^p_i is the neighbor service set of service A_i under meta-path p, and e^p_i is the node level embedding embeddings of service A_j under meta-path p.

Due to the scale-free property of the heterogeneous graph, the variance of graph data is large. To stabilize the learning process of the self-attention mechanism, we use the multi-head attention, repeats the above calculation several times to obtain multiple representations of the service in a single meta-path, and then use concatenation to obtain the representations of K times. It is shown in Formula (7).

$$\boldsymbol{e'_i} = ||^{\boldsymbol{K}}_{\boldsymbol{k}=1}(\boldsymbol{e_i}) \tag{7}$$

Finally, for the whole dataset, we describe the final service node embeddings matrix at the node level under the meta-path as $\Phi_p = \{e'_1, e'_2, \ldots, e'_N\}$.

3) Meta-Path-Level Attention: Through the calculation at the node level, we get the embedding matrix of services under each meta-path. To integrate meta-paths to learn more comprehensive node embedding, we use a new attention mechanism to automatically learn the importance of different meta-paths. First, we use a layer of MLP to perform nonlinear transformation for each node level embeddings result. For the transformed embedding vector, the meta-path-level attention is measured as the similarity between the transformed node level embedding vector and the attention parameter vector. Obviously, the higher the attention weight is, the more important the meta-path is. It is shown in Formula (8).

$$a_{met}{}^p_i = soft\max\left(\frac{1}{|V|}\sum_{i \in V} \theta^T \bullet \tanh(W_{met} \bullet e^p_i + b)\right) \tag{8}$$

Among them, θ^T is the semantic attention vector, W_{met} is the parameter matrix of the MLP layer, and b is a bias vector. Similarly, all the above parameters are shared for each attention calculation. V is all service nodes, and $a_{\text{met}}{}^p_i$ is the attention weight of service A_i under the p-th meta-path. After getting the importance of each meta-path, we normalize them by Softmax function.

Taking the learned weight as the coefficient, we can integrate the service embedding calculated at the node level to get the final embedding result. It is shown in Formula (9).

$$Z_i = \sum_{p \in P} a_{met}{}^p_i \bullet e'^p_i \tag{9}$$

Among them, e'^p_i is the node embeddings of service A_i under meta-path p after node level attention calculation; Z_i is the result of aggregating each meta-path representation.

4) Service Category Prediction: Finally, the service is embedded into a full connection layer, and the Softmax activation function is used to output the probability distribution of all candidate Web service categories. Softmax transforms the output value of multi-classification into relative probability, which represents the probability that the node belongs to a certain category. Finally, for semi-supervised node classification, we minimize the cross-entropy of all service nodes between the real label and the predicted label to calculate the loss value. It can be noted as Formula (10):

$$L = -\sum_i Y'_i log(Y_i) \tag{10}$$

Where Y'_i refers to the ith value in the actual label, Y_i refers to the value of the ith element predicted after Softmax. Under the guidance of labeled data, the embedded nodes are learned by the back-propagation optimization model.

C. Score Prediction and Ranking Based on FiBiNET

By using the FiBiNET, the discrete and continuous features of service are used as input of a multidimensional matrix, and the score prediction is performed to predict the score value of each API to the target Mashup. In the recommendation system, the combination of different features in the data have a great impact on the recommendation results. The traditional CTR model is the basis of the deep learning CTR model. For example, the influential FNN [32], DeepFM [33], xDeepFM [34], and other deep learning models are inextricably linked with the traditional FM model. At present, many works of CTR prediction through feature combination mainly use the inner

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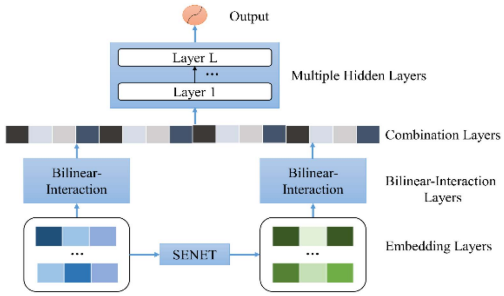

Fig. 5. The overall framework of FiBiNET.

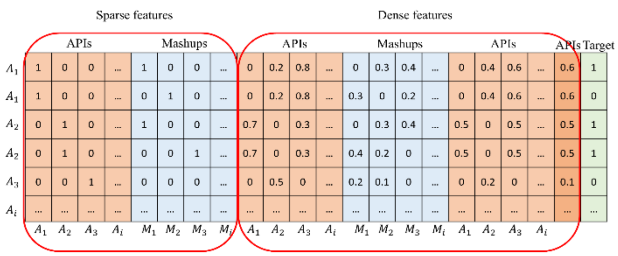

Fig. 6. Multi-dimensional information matrix.

product or Hadamard product of feature vector to calculate the cross features. This method ignores the importance of the features themselves, and it is difficult to learn the important features that never appear or rarely appear from the training samples [13]. FiBiNET dynamically learns the importance of features by using the SENET [35] structure and uses a bilinear function to better model cross features. As shown in the Fig. 5, the original embedding and the new embedding with SENET adjusted weight learn the second-order interaction features at the bilinear-interaction layer, and then further learn the high-order features through MLP after they are spliced.

1) Multi-Dimensional Information Matrix: Using the FiBiNET, the multidimensional matrix constructed from the coding, neighbor similarity, co-occurrence, and popularity of Web APIs and Mashups can be used as input to predict the score value of each API to the target Mashup, as shown in Fig. 6 below:

The discrete features mainly include two parts:

1) *Coding matrix of Web API:* mark the serial number of Web API, and map each Web API into a unique one-hot code.
2) *Mashup coding matrix:* mark the number of Mashups, and map each Mashup into a unique one-hot code.

The continuous feature mainly consists of four parts:

1) *Similarity matrix of Web APIs:* the heterogeneous graph attention network is used to calculate the service embeddings at the node level and meta-path-level, and then the cosine similarity calculation formula is used to calculate the similarity between Web APIs.
2) *Similarity matrix of Mashups:* use heterogeneous graph attention network to calculate the service embeddings of Mashups at the node-level and meta-path-level, and then use the cosine similarity formula to calculate the similarity between Mashups.
3) *Co-occurrence matrix of Web APIs:* represent the historical combination relationship of Web APIs. If two Web APIs have been called by a Mashup at the same time, they are co-occurrence.
4) *Popularity of Web API:* reflects the popularity of Web API. The higher the popularity, the easier it is to call the Web API. The calculation of popularity is shown in Formula (11), where $\mathrm{F}(*)$ represents the number of times an API is called by Mashup, and $\mathrm{C}(*)$ represents all APIs under the category of the API.

$$pop = \frac{F\left(API_i\right) - \min\left(F\left(C\left(API_i\right)\right)\right)}{\max\left(F\left(C\left(API_i\right)\right)\right) - \min\left(F\left(C\left(API_i\right)\right)\right)} \tag{11}$$

2) SENET Layer: The original input features are sparse. First, the sparse and high-dimensional features are embedded into low-dimensional and continuous vectors through the embedding layer. The simple use of embedded features after linear transformation ignores the importance of different features. For example, if the popularity of service is more important to user than the co-occurrence of service, it will gain more bigger weight, and vice versa. SENET mechanism can deal with this problem well. It can better model feature intersection and enrich the representation of original sparse data. And it can also dynamically learn the importance of the features, weaken the low frequency unreliable features, and strengthen the high frequency features. Through this mechanism, we can dynamically increase the weight of important features and reduce the importance of features with insufficient information [35]. This is also similar to the attention mechanism [36] in the previous article. The difference is that the use of the attention mechanism in the previous article pays more attention to the relationship between nodes, while the SENET pays more attention to the relationship between features.

Therefore, here we generate SENET-like embedding after selecting the importance of the features after linear transformation through SENET layer. The features are compressed by averaging pooling and feature excitation is performed. Here we use two layers of MLP, the first layer is used to reduce the dimension and the second layer is used to restore the dimension. For all current input features, we can dynamically judge which features are important and which are not important by associating them with each other. In this way, we can learn the relationship between various features and get their weights.

$$A = \sigma_2\left(W_2\sigma_1\left(W_1\left(\frac{1}{k}\sum_{t=1}^{k} e^{(t)}\right)\right)\right) \tag{12}$$

Where e is the feature after linear transformation, k is the embedded dimension, σ_1 and σ_2 are nonlinear functions. Here we use relu function, and W_1 and W_2 are the parameter matrices of two fully connected layers.

$$V = [a_1 \cdot e_1, \ldots, a_f \cdot e_f] = [v_1, \ldots, v_f] \tag{13}$$

Among them, V is the SENET-like embedding result.

3) Bilinear-Interaction Layer: After feature importance selection, feature interaction is carried out next. In previous models, inner product (such as FM, FFM) or Hadamard product (such

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as NFM, AFM) are often used for feature interaction. Using the inner product or Hadamard product alone cannot consider the importance of each dimension of the vector. Therefore, we consider using bilinear-interaction, that is, inner product first and then Hadamard product. Bilinear operation is a more fine-grained method, which considers the important information of each dimension of the interactive vector at the same time. We conduct bilinear-interaction between the features of services after linear transformation and the features after senet processing, as shown in Formula (14).

$$p_{ij} = v_i \cdot W_{ij} \odot v_j \tag{14}$$

Where p_{ij} is the interaction vector, and $W_{ij} \in \mathbb{R}^{k \times k}$ corresponding to the connection/weight matrix between the *i*-th feature of service and the *j*-th feature of service. The original embedded E output interaction vector is p, and the SENET-like embedded V output interaction vector is q.

4) Predict Layer: We connect the interaction vector p and q, to capture the high-order implicit interaction through several full connection layers, finally generate a dense real-valued vector, and use the sigmoid function to predict:

$$\hat{y} = sigmoid\left(w_0 + \sum_{i=0}^{m} w_i x_i + y_d\right) \tag{15}$$

Where, y_d is the output after deep neural network, w_i is the *i*-th weight of the linear part, and $\hat{y}$ is the final prediction result.

D. Web Service Recommendation

Web service recommendation is divided into below steps:

1) The heterogeneous service information network is constructed, the meta-path is extracted based on semantics, and the commuting matrix is calculated by PathSim, to obtain the service adjacency matrix.
2) The heterogeneous graph attention network is used to calculate the embeddings of node level and meta-path-level respectively and fuse them. The results are divided into service categories by Softmax.
3) The embedding results are combined with the service score to form service information matrix, the feature importance is selected by SENET, the feature interaction is carried out through bilinear-interaction layer, and the service is recommended according to the final score prediction results.

In fact, the final score prediction result is a continuous value. When the score is greater than or equal to 0.5, the corresponding Web API is recommended, and otherwise, the corresponding Web API is not recommended.

V Experimental Evaluation and Analysis

A. Dataset and Experimental Settings

We use the dataset crawled from the ProgrammableWeb to conduct the experiment. This dataset contains 22622 Web APIs (referred to as character A), 6467 Mashups (referred to as character M). For each Web API, the information includes its name, description text, category, tag information, and so on. For each Mashup, the information includes its name, description text, category, label, and the invoked Web API. We also develop an Service Ecosystem Platform based on the dataset, includes Service Recommendation on the URL https://apirecommendhnust.ingress.isa.buaanlsde.cn/, and Service Network on the URL https://apinetworkhnust.ingress.isa.buaanlsde.cn/.

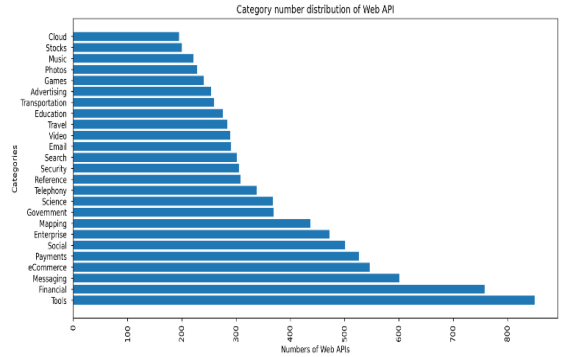

Fig. 7. Number distribution of Web APIs in the top 25 categories.

TABLE III
The Statistics of Web Services Dataset

Items	Values
Number of Web APIs	22622
Number of Mashups	6467
Number of APIs (per Mashup)	3.50
The average number of feature words (per API)	41
The average number of feature words (per Mashup)	24.1

The experiments are performed on the basic configurations, which include Intel(R) Xeon(R) Silver 4210 CPU@2.20 GHz, 64 G memory, Nvidia 2080Ti 11GB video card, and 2 TB hard disks. There are 400 Web APIs categories in the dataset, and we select the top 5, 10, 15, 20, and 25 categories with the largest number of APIs as the experimental objects. The number of Web APIs in the top 25 categories already account for about 80% of the entire dataset, and their distribution is shown in Fig. 7. Some of the relevant statistics for the Web services dataset are shown in Table III. To make the results more reliable, the experimental data is randomly scrambled during the training, and then 60% of the data are divided into training sets, 20% of the data are divided into verification sets, and 20% of the data are divided into test sets for multiple experiments. Adam [37] is used as the optimizer of the model. Among this, in the testing process, the embeddings of the mashups are generated only in terms of their descriptions, the values of other feature informations are all set to 0. That is to say, as for the multi-dimensional information matrix in the Fig. 6, mashup coding is a one-hot code and similarity of Mashups is the cosine similarity between Mashups derived from HGAN, while the coding, similarity, co-occurrence, and popularity of Web API are all 0.

B. Web API Classification Experiment and Analysis

1) Evaluation Metrics: In this paper, Recall (R_i), Precision (P_i) and Macro-F1 (F) are used as evaluation indexes to evaluate

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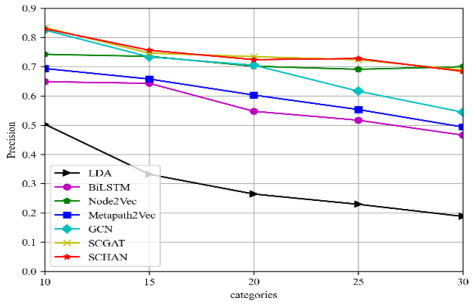
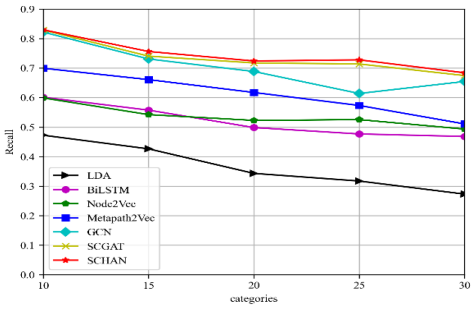
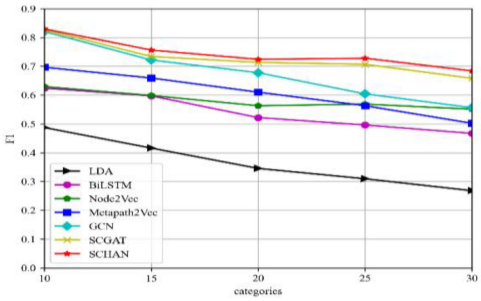


(a) Precision



(b) Recall



(c) Macro-F1

Fig. 8. Results of service classification.

the performance of the above methods for Web service classification. Recall describes the proportion of correctly classified Web APIs in all Web APIs of this category. Precision describes the proportion of Web APIs that really belong to this category in the final classification results of the model. F1 is the harmonic average of recall and accuracy. Their computation formulas are as follows:

$$R = \frac{\sum_i^N R_i}{N} \tag{16}$$

$$P = \frac{\sum_i^N P_i}{N} \tag{17}$$

$$F = \frac{2 \times R \times P}{R + P} \tag{18}$$

2) Baselines:

- *LDA [38]:* an unsupervised machine learning method, which can be used to identify topic hidden in large-scale document sets or corpora. Each document is regarded as a word frequency vector by using the method of word bag, so the text information is transformed into digital information which is easy to model.
- *BiLSTM [39]:* it is composed of forward and backward LSTMs. The last hidden vector in BiLSTM is taken as the embeddings of service document, and the probability distribution of category is obtained by Softmax.
- *Node2Vec [40]:* the sequence corresponding to each point is obtained by using a specific walk mode, and two parameters are defined to balance the influence of BFS and DFS. At the same time, the local and global information of graph structure is considered, which has high adaptability.
- *Metapath2Vec [41]:* use the random walk based on metapath to obtain the heterogeneous domain of each different type of vertex in the heterogeneous network, then use the extended skip-gram to process the vertex domain. Here, we test all meta-paths with metapath2Vec and get the best results.
- *GCN [20]:* GCN takes the standardized graph structure and node features as input, and uses the eigenvalues and eigenvectors of the Laplacian matrix of the graph to extract the spatial features of the topological graph. Here, we test all meta-paths with metapath2vec and get the best results.
- *SCGAT [10]:* through the attention mechanism to aggregate the neighbor nodes, realize the adaptive allocation of different neighbor weights, thus greatly improving the expression ability of the graph neural network model.

TABLE IV
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON MACRO-F1

Method	Category				
	10	*15*	*20*	*25*	*30*
LDA	0.4874	0.4161	0.3455	0.3096	0.2681
BiLSTM	0.6241	0.5973	0.5219	0.4961	0.4671
Node2Vec	0.6849	0.6510	0.6019	0.5829	0.5684
Metpath2Vec	0.6966	0.6594	0.6101	0.5632	0.5021
GCN	0.8207	0.7217	0.6777	0.6042	0.5563
SCGAT	0.8271	0.7341	0.7134	0.7069	0.6583
SCHAN	**0.8298**	**0.7463**	**0.7241**	**0.7280**	**0.6840**

3) Experimental Results and Analysis: In the experiment, we repeatedly train the experimental data for three times. When the deviation of the experimental results is less than 0.01, we take the average value of the three experimental results. When the deviation of the experimental results is large, we readjust the model and observe the experimental data to achieve a more stable result. Among this, the average Macro-F1 value as shown in Table IV. It can be seen from Table IV that the method based on heterogeneous graph attention has better performance than other methods. Especially when the number of categories is 25, SCHAN increases by 41.84%, 23.19%, 14.51%, 16.48%, 12.38%, and 2.11% respectively on Macro-F1 compared with other baselines. Specifically, LDA and BiLSTM perform the worst because of sparse features and insufficient consideration of the relationship between services and other objects. Node2vec and metapath2vec are relatively better, which indicates the importance of considering the relationship extraction of services.

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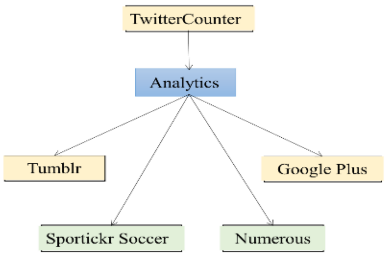
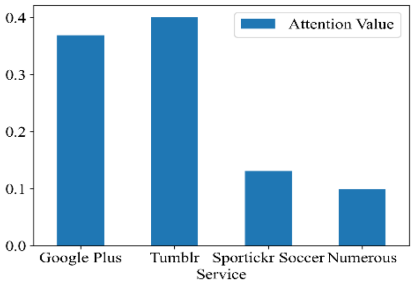


(a) neighbors of "TwitterCounter"



(b) attention values

Fig. 9. Attention weight of neighbor services.

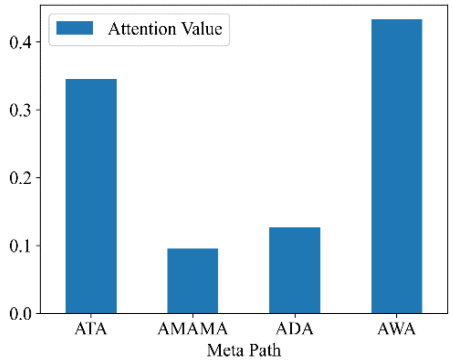


Fig. 10. Attention weight of meta-path.

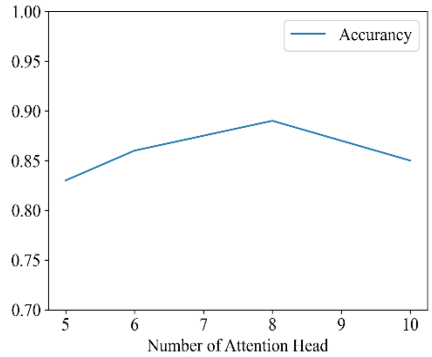


Fig. 11. The influence of attention weight on accuracy.

For the same attention calculation, the method based on heterogeneous graph attention network is better than graph attention network which only considers node level attention, which shows that the method proposed successfully captures the rich semantics in the meta-path and shows its superiority. Through the above analysis, we can find that our method shows good performance. The results show that it is very important to integrate HIN and attention mechanism to analyze the importance of nodes and meta-paths.

4) Hyper-Parameter Analysis. Attention Weight Analysis of Neighbor Services: Taking "TwitterCounter" as an example, we list its neighbors based on the meta-path "ATA", and their attention values after model training are shown in Fig. 9. As can be seen from Fig. 9(a), "TwitterCounter" is linked to "Google plus" and "Tumblr" through a meta-path, both of which belong to the category of "social"; while the linked services "Sportickr Soccer" and "Numerous" belong to other categories. As can be seen from Fig. 9(b), "Google plus" and "Tumblr" get higher attention values through node level attention calculation, and the weight of the impact on the classification of "TwitterCounter" is 0.369 and 0.401 respectively, which means that service nodes get more weight are more useful for service classification. Meanwhile, "Sportickr Soccer" and "Numeric" get less attention, and the weight of the impact on the classification of "TwitterCounter" is 0.131 and 0.099 respectively, which is reasonable because they do not play an important role in the classification of "TwitterCounter". Thus, we can conclude that node level attention distinguishes the differences between neighbors, and gives higher weight to meaningful neighbors.

Attention Weight Analysis of Meta-Path: To verify the effect of meta-path-level attention, we show the final corresponding attention value of each meta-path, as shown in Fig. 10. For meta-path "ATA", "AMAMA", "ADA", and "AWA", the meta-path-level attention mechanism gives the path "ATA" the largest weight, which means that meta-path "ATA" is the most helpful meta-path for Web service classification. This is reasonable because the service function category and its function label are highly consistent. At the same time, it can be found from the Fig. 10, that the importance weight of meta-path "AMAMA" is very low. This is because although there are many services, not everyone interacts with Mashups with a high frequency and lacks the useful effect on service classification to a certain extent. If we treat these meta-paths equally, we will not be able to deal with the complex relationships in the service relationship network well. Through the above analysis, it can be concluded that there is a positive correlation between the performance of single meta-path and its attention value.

Number of Attention Heads K: To test the effect of multiple attention, we study the performance of Web services classification when the number of attention heads varies. It can be seen from Fig. 11 that increasing the number of attention heads can effectively improve the experimental results within a certain range. When the number of attention heads increases from 2 to 8, the classification accuracy is improved to a certain extent. Therefore, the number of attention heads should not be too small, otherwise, the experimental effect will be reduced to a certain extent. With the increase of the number of attention heads, the calculation amount of the attention layer increases, which makes the training time longer. When the number of attention heads

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reaches 10, the training speed is relatively slower, which is slightly lower than the experimental results when the number of attention heads is 8. Therefore, it is suggested that the number of attention heads equals 8. However, we can infer that when the number of attention heads is too large, it will not only increase the computational cost, but also the repeated calculation of attention will lead to information redundancy and make the test results worse.

C. Web API Recommendation Experiment and Analysis

1) Evaluation Metrics: Based on the classification results, we perform the recommendation experiment. In this part, AUC_ROC, Logloss, Recall, and F1 are used as evaluation metrics. AUC is the area under the ROC curve, which is widely used in CTR prediction and is a good measurement method. When the value of AUC is closer to 1.0, its performance effect is better. Logloss is a widely used measure in binary classification. It measures the accuracy of the classifier by punishing the wrong classification. The minimum value of the logarithmic loss is 0. At this time, it means that the predicted results are completely matched with the real results. The smaller the value, the better the performance.

2) Baselines: Here, based on the two existing best classification methods SCGAT and SCHAN, the combined recommendation experiments are compared with FNN, DCN, DeepFM, and FiBiNET respectively, as follows:

- *SCGAT-FNN:* Based on the representations achieved by SCGAT, FNN [32] is used for Web API recommendation. FNN (factorization machine supported neural network) uses FM for supervised learning. The embedding vector of each feature is trained by the FM model in advance. When learning the DNN model, the training overhead is reduced and the convergence is fast.
- *SCGAT-DCN:* Based on the representations achieved by SCGAT, DCN [42] is used for Web API recommendation. DCN uses cross network and applies feature crossing at each layer. Thus, the combined features of bounded degree can be learned efficiently without artificial feature engineering.
- *SCGAT-DeepFM:* Based on the representations achieved by SCGAT, it uses DeepFM [33] to recommend API. DeepFM combines linear crossover of FM to second-order features with the nonlinear crossover of neural network to higher-order features and can learn the feature combination that has not appeared in the training set.
- *SCGAT-FiBiNET:* It uses FiBiNET [13] to recommend API based on the representations achieved by SCGAT. FiBiNET adds a SENET layer to the traditional embedding stage, embeds the features that have been embedded, and then uses a new bilinear-interaction method to obtain the relationship between the features together with the original embedding feature.
- *SCHAN-FNN:* Based on the representations achieved by SCHAN, FNN is used for Web API recommendation. FNN ditto.

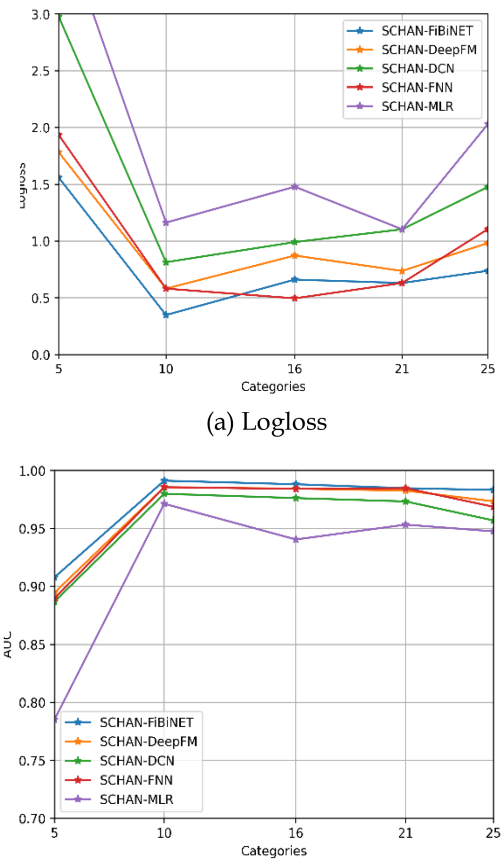

Fig. 12. Recommendation effect of SCHAN-Fi under different Mashup categories.

- *SCHAN-DCN:* Based on the representations achieved by SCHAN, DCN is used for Web API recommendation. DCN ditto.
- *SCHAN-DeepFM:* Based on the representations achieved by SCHAN, DeepFM is used for Web API recommendation. DeepFM ditto.
- *SCHAN-FiBiNET:* Based on the representations achieved by SCHAN, FiBiNET is used for Web API recommendation. FiBiNET ditto.

3) Experimental Results and Analysis: During the experiment, we investigated the effect of Web API recommendation under different number of Mashup services (i.e., top-N = 5/10/15/20/25). As can be seen from Fig. 12, overall, the service recommendation effect is the best when the number of Mashup categories is 10. This may be because there are few relationships between services that can be mined under category 5, while category 10 includes most services with high usage data, which is easy to mine rich semantic information. With the increase of the number of classifications, the QoS data of some services is difficult to support the model for information mining.

At the same time, when the number of Mashups is 5, due to the impact of function categories and the number of services, the proportion of calls from Mashups to Web APIs is only 0.294. After 10 classification, we control the proportion of

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TABLE V
PERFORMANCE COMPARISON OF WEB API RECOMMENDATION UNDER 10 CLASSIFICATIONS

Methods		Logloss	AUC	Recall	F1
SCGAT	MLR	1.2899	0.9354	0.9189	0.9142
	FNN	1.0490	0.9557	0.9374	0.9496
	DCN	0.9454	0.9601	0.9452	0.9681
	DeepFM	0.8546	0.9612	0.9545	0.9644
	FiBiNET	0.6590	0.9751	0.9684	0.9754
SCHAN	MLR	1.1629	0.9713	0.9425	0.9704
	FNN	0.8140	0.9799	0.9598	0.9795
	DCN	0.5815	0.9856	0.9713	0.9854
	DeepFM	0.5815	0.9856	0.9713	0.9854
	FiBiNET	**0.3489**	**0.9912**	**0.9825**	**0.9912**

Mashups calling Web APIs in the samples to be about 0.5, and the proportion of positive and negative samples is relatively balanced. When the number of Mashup categories exceeds 10, although the number of services increases, the richness of samples also increases. When performing feature crossing, the number of samples cannot keep up with the change of the variety richness, so the recommendation effect decreases. Therefore, we selected the method with 10 Mashup categories for performance comparison with the method based on SCGAT.

When the number of service categories is 10, we compare it with other methods. Table V shows the performance comparison results of different recommended methods, including Logloss, AUC, Recall, and F1. The performance of our method is significantly better than other methods. Specifically,

- The performances of all Web API recommendation methods based on classification are all good. For example, the AUC, Recall and F1 of all Web API recommendation methods based on SCGAT and SCHAN greater than 0.9. This indicates that Web API classification in advance can identify the Web APIs with similar functionality, which can greatly facilitate Web API recommendation.
- All methods based on SCHAN are generally higher in AUC than all methods based on SCGAT. This is because when choosing the representation learning model, we make further improvements based on SCGAT, and use SCHAN to extract meta-path semantics by constructing heterogeneous service information network. Compared with SCGAT, which integrates the relationship between services and other nodes into a matrix, we use the meta-path to define the adjacency matrix and further calculate the node level embeddings to the meta-path-level embeddings, which can better deal with various types of nodes and edges and capture richer semantics. This is also verified in the Web service classification task.
- A series of methods based on FiBiNET achieves the best performance. The results show it is effective to combine SENET mechanism with bilinear-interaction in DNN, and implicit high-order feature interaction is helpful to obtain greater expression ability of shallow model.

VI CONCLUSION AND FUTURE WORK

Based on the service classification generated from hierarchical attention model, combined with service prediction based on multidimensional service quality information, this paper proposes a Web service recommendation method integrating heterogeneous graph attention network representation and FiBiNET score prediction. The heterogeneous service information network is modeled by using the service network with rich attributes and relationships. Second, the meta-path with important semantics is extracted, and the PathSim is used to calculate the service similarity matrix. Then, the neighbor information is aggregated based on the node level attention mechanism, and the meta-path information is aggregated through the meta-path-level attention mechanism, considering the importance of neighbors and meta-paths. Finally, combined with multi-dimensional QoS attribute information, the FiBiNET model is used to dynamically learn the importance of features and complex feature interaction, to recommend high-quality Web services for developers. The experimental results show that compared with other typical methods, the proposed method has achieved better classification and recommendation results. This paper integrates service content information and service quality information to carry out service recommendation, and mainly makes relevant exploration of service content. In the future, we can also mine the relevant technologies of service quality information to comprehensively improve the quality of service recommendations.

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